



Analysis of Bridges & Wormholes Paper

Part 3 – Consider Data Handling, Shortcomings and Improvements

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Michael M Lee



Introduction

Objective of Study

- Provide an overview of the B&W approach that makes it more accessible

Approach to Study

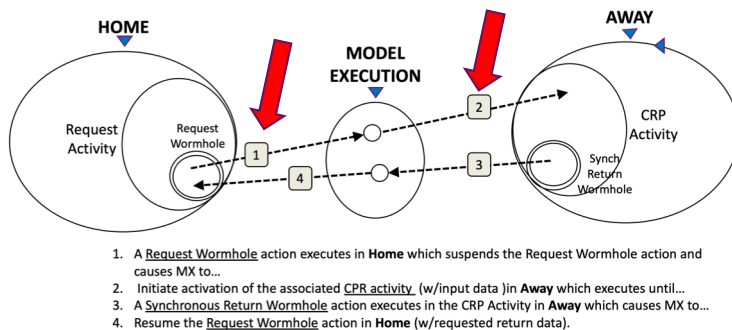
- Understand and explain the proposed wormhole approach to bridging
- Assess if the responsibilities of the SW Architecture proposed in the B&W paper are sufficient and feasible for the proposed approach
- Identify shortcomings and propose improvements
- Separate the study into considering:
 - Control ✓
 - ➔ • Data Handling

Approach to Characterizing Data Handling



Synchronous Bridge Scenario

Simplified View of a Synchronous Bridge Scenario

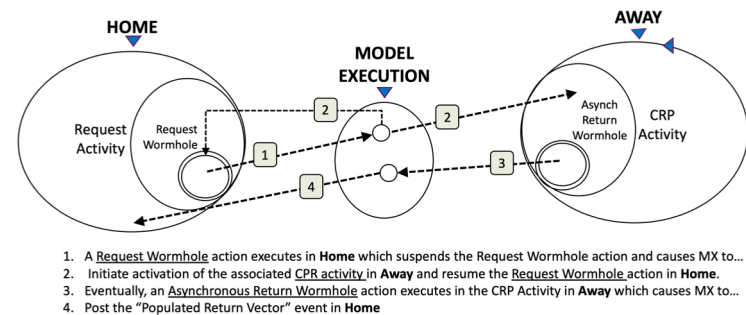


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Asynchronous Bridge Scenario

Simplified View of an Asynchronous Bridge Scenario



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1. Use the sequencing of the Control study as a context
2. Identify the Data Handling required at each step of the sequence
3. Start with steps 1 & 2 of synchronous bridge scenario



But First... Fix the Information Models

Both the HOME and AWAY domains recognize that they have “setpoint devices” that have target and measured settings as evidenced by their request wormhole and synchronous service:

HOME

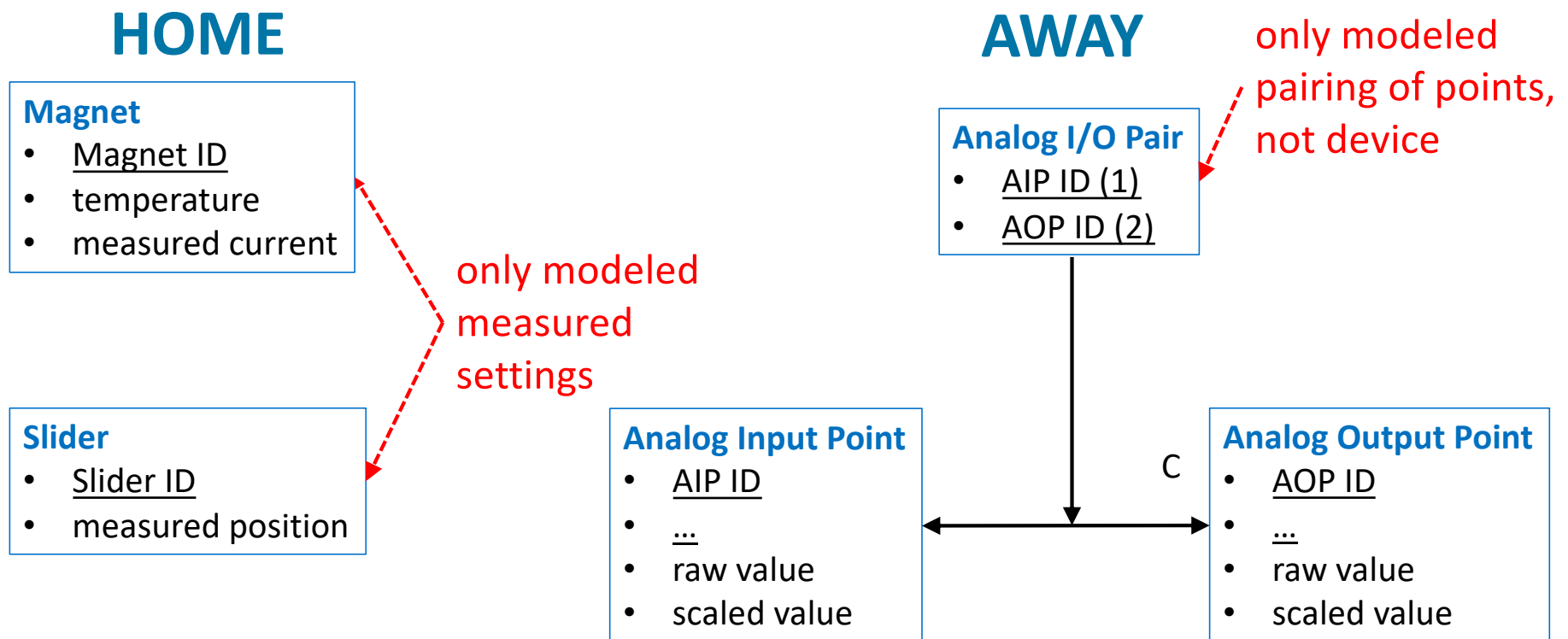
Request Wormhole: Position Slider (in: Slider ID, **target position**; out: **delta**)

AWAY

CRP: Iterate to Setpoint (in: AIP ID, **setpoint**, return coordinate; out: **discrepancy**)

But HOME only models the target setting and AWAY only recognizes the pairing of I/O points for a setpoint device, not that the device actually exists

Initial Information Model Classes





Revised Information Model Clases

HOME

Magnet

- Magnet ID
- temperature
- target current
- measured current

Slider

- Slider ID
- target position
- measured position

modeled target
and measured
settings

AWAY

Setpoint Device

- SN (1)
- AIP ID (2)
- AOP ID (3)

modeled device
with paired
points

Analog Input Point

- AIP ID
- ...
- scaled value

Analog Output Point

- AOP ID
- ...
- scaled value

C

Handling Data when Constructing the Bridge



Prescription for Construction from the Bridges and Wormholes Paper

- 
1. Populate the participating domains
 2. Match each Wormhole with a Control Reception Point
 3. Develop correspondences between data values that cross the domain boundary



Original AWAY Signal Lists

AWAY

Analog Input Point			
AIP ID	Register	a	b
1	1	.3	0
2	2	.1	0
3	3	.5	0
4	4	.1	0
...	...	...	...
48	75	1	.03
...	...	...	...
96	123	10	0
97	124	.5	0
98	125	.36	-180.
...	...	...	...
114	33	1	0
115	34	1	0
116	35	1	0
...	...	...	...

HOME

magnet 1 current
magnet 1 temperature
magnet 2 current
magnet 2 temperature

slider

robot 1 x-axis
robot 1 y-axis
robot 1 θ -axis

field probe x-axis
field probe y-axis
field probe z axis

Analog Output Point			
AOP ID	Register	c	d
1	21	3.3	0
2	22	2	0
3	23	1.67	0
...	...	...	...
17	202	.1	0
18	203	2.	0
19	203	2.78	500
20	143	1	0
21	144	1	0
22	146	1	0
...	...	...	...
31	13		
...	...	...	...

magnet 1 current
magnet 2 current
magnet 3 current

...

robot 1 x-axis
robot 1 y axis
robot 1 θ axis

field probe x-axis
field probe y-axis
field probe z axis

slider position
...

What's going
on here?

AWAY

Analog Input Output Pair	
AIP ID	AOP ID
1	1
3	2
5	3
...	...
48	31
...	...
96	17
97	18
98	19
...	...
114	20
115	21
116	22
...	...

HOME

magnet 1 current
magnet 2 current
magnet 3 current

slider

robot 1 x axis
robot 1 y axis
robot 1 θ axis

field probe x axis
field probe y axis
field probe z axis

New AWAY Signal Lists



Analog Input Point

AIP ID	HOME Class	HOME Class ID	HOME Attribute
1	Magnet	1	measured current
2	Magnet	1	temperature
3	Magnet	2	measured current
4	Magnet	2	temperature
5	Magnet	3	measured current
6	Magnet	3	temperature
7	Slider	1	measured position
8	Slider	2	measured position
9	Slider	3	measured position

AWAY

HOME

Analog Output Point

AOP ID	HOME Class	HOME Class ID	HOME Attribute
1	Magnet	1	target current
2	Magnet	2	target current
3	Magnet	3	target current
4	Slider	1	target position
5	Slider	2	target position
6	Slider	3	target position

Setpoint Device

SN	AIP	AOP	HOME Class	HOME Class ID
100	1	1	Magnet	1
101	3	2	Magnet	2
102	5	3	Magnet	3
1000	7	4	Slider	1
1001	8	5	Slider	2
1002	9	6	Slider	3

Handling Data when Constructing the Bridge



Prescription for Construction from the Bridges and Wormholes Paper

1. Populate the participating domains
2. Match each Wormhole with a Control Reception Point
3. Develop correspondences between data values that cross the domain boundary



Matched Wormholes and Control Reception Points



Wormholes

CRPs

- W1 Get Magnet Current → CRP1 Read AIP
 - W2 Get Magnet Temperature → CRP1 Read AIP
 - W3 Get Slider Position → CRP1 Read AIP
 - W4 Set Slider Position → CRP2 Iterate to Setpoint
 - W5 Set Magnet Current → CRP2 Iterate to Setpoint
- Same CRP Service
- Same CRP Service

Handling Data when Constructing the Bridge



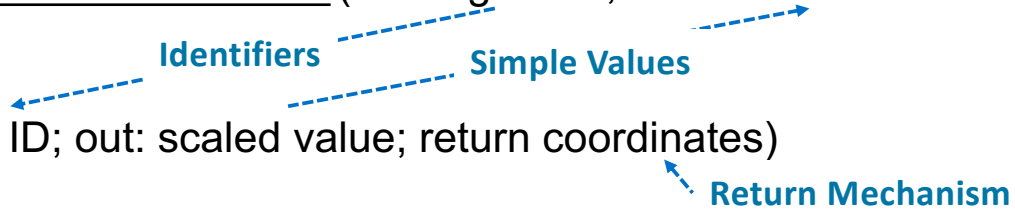
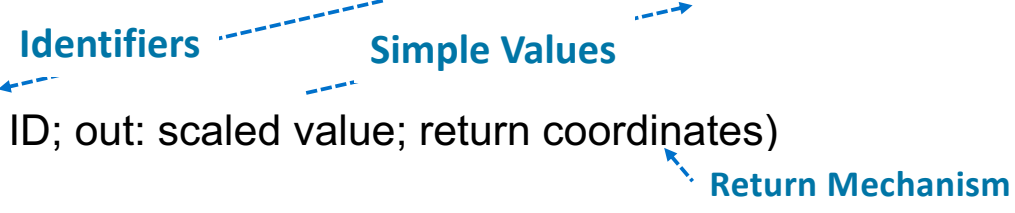
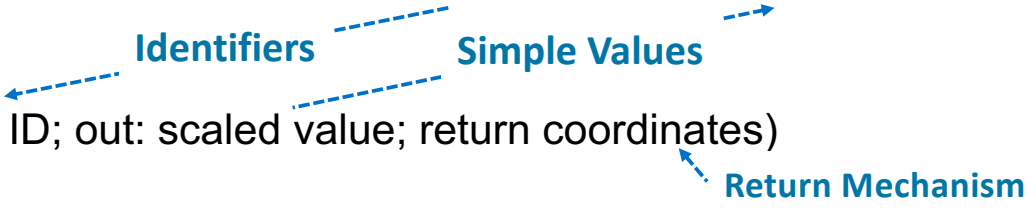
Prescription for Construction from the Bridges and Wormholes Paper

1. Populate the participating domains
2. Match each Wormhole with a Control Reception Point
3. Develop correspondences between data items that cross the domain boundary





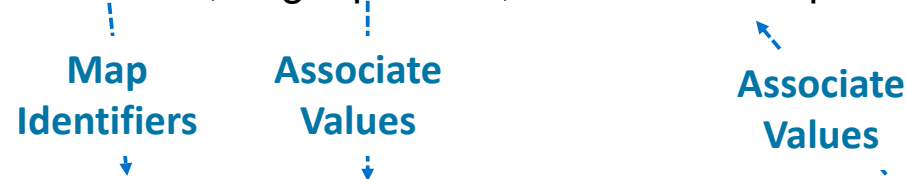
Types of Corresponding Data Items

Request	W1-Get <u>Magnet Measured Current</u> (in: Magnet ID; out: measured current)
CRP	Read AIP (in: AIP ID; out: scaled value; return coordinates)
	
Request	W2-Get <u>Magnet Temperature</u> (in: Magnet ID; out: temperature)
CRP	Read AIP (in: AIP ID; out: scaled value; return coordinates)
	
Request	W3-Get <u>Slider Measured Position</u> (in: Slider ID; out: measured position)
CRP	Read AIP (in: AIP ID; out: scaled value; return coordinates)
	



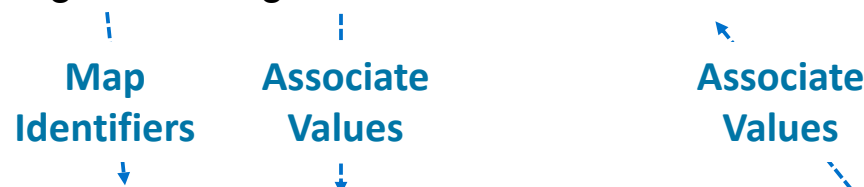
Types of Corresponding Data Items

Request: W3-Position Slider (in: Slider ID, target position; out: measured position)



CRP: SETD3-Iterate to Setpoint (in: SN, AOP.scaled value; out: AIP.scaled value, return coord)

Request: W30-Set Current (in: Magnet ID, target current; out: measured current)



CRP: SETD3-Iterate to Setpoint (in: SN, AOP.scaled value; out: AIP.scaled value, return coord)



Different Data Handling Required for Each Type of Data Item

1. Identifiers → Need to be mapped.
2. Simple Values → Need to be matched between the correct data items in the respective interfaces and may need type and semantic conversions.
3. Return Mechanism → Are managed by the software architecture/model execution domain. They are only stored and retrieved by application models.



Data Handling for Identifiers

Identifier mappings can be specified in “half tables” at the Request level and implemented with generic code.

**Get Magnet
Measured Current**

Mag ID	AIP ID
1	1
2	3
3	5

**Get Magnet
Temperature**

Mag ID	AIP ID
1	2
2	4
3	6

**Get Slider
Measured Position**

Slider ID	AIP ID
1	7
2	8
3	9

NOTE: This does not address the issue of maintaining these tables when the instance population is dynamic.



Half Table Scope: Domains versus Request

Application Domain					PIO Domain		
object name ⁴	identifier attribute name	identifier attribute value	non-identifying attribute name		object name ⁴	identifier attribute name	identifier attribute value
Magnet	Magnet ID	1	current		AIP	AIP ID	1
Magnet	Magnet ID	1	current		AOP	AOP ID	1
Magnet	Magnet ID	1	temperature		AIP	AIP ID	2
Magnet	Magnet ID	2	current		AIP	AIP ID	3
Magnet	Magnet ID	2	current		AOP	AOP ID	2
Magnet	Magnet ID	2	temperature		AIP	AIP ID	4
Magnet	Magnet ID	3	current		AIP	AIP ID	5
Magnet	Magnet ID	3	current		AOP	AOP ID	3
Magnet	Magnet ID	3	temperature		AIP	AIP ID	6
...	...	...	...		...	...	...
RH Robot	Robot ID	1	x		AIP	AIP ID	96
RH Robot	Robot ID	1	x		AOP	AOP ID	17
RH Robot	Robot ID	1	y		AIP	AIP ID	97
RH Robot	Robot ID	1	y		AOP	AOP ID	18
RH Robot	Robot ID	1	θ		AIP	AIP ID	98
RH Robot	Robot ID	1	θ		AOP	AOP ID	19
...	...	...	...		...	...	...
Field probe	Field probe ID	1	x		AIP	AIP ID	114
Field probe	Field probe ID	1	x		AOP	AOP ID	20
Field probe	Field probe ID	1	y		AIP	AIP ID	115
Field probe	Field probe ID	1	y		AOP	AOP ID	21
Field probe	Field probe ID	1	z		AIP	AIP ID	116
Field probe	Field probe ID	1	z		AOP	AOP ID	21
...	...	...	...		...	...	...
Slider	Slider ID	1	position		AIP	AIP ID	48
Slider	Slider ID	1	position		AOP	AOP ID	31
...	...	...	...		...	...	...

Large and Complex

VS

Get Magnet Measured Current

Mag ID	AIP ID
1	1
2	3
3	5

Get Magnet Temperature

Mag ID	AIP ID
1	2
2	4
3	6

Smaller and Simpler

Data Handling for Simple Values



Matching Simple Values between the correct data items in the wormhole and CRP interfaces is a straightforward programming language capability and can readily be done by the Model Execution domain when handling a wormhole invocation.

Type conversions for Simple Value data items is a straightforward programming language capability and can readily be done by the Model Execution domain when handling a wormhole invocation.

Semantic conversions for Simple Value data items, however are not addressed in the B&W paper. The one case where it could have been required (scaling the raw data of an AIP read to engineering units) the semantic conversion logic was put in the AWAY domain and the issue avoided. If a conversion was required, and it was not conveniently achieved with generic, data driven code, it is unclear where the bridge specific logic would go, since the B&W paper makes no accommodations for adding additional logic to the bridge.

Data Handling for Return Mechanisms



The B&W approach introduced two Return Mechanism data items:

- Return Coordinates and
- Transfer Vector

Return Coordinates are created and used by the software architecture /model execution domain. The modeled domains have to store and retrieve them as an opaque data structure. These can be readily done.

Transfer Vectors are also created and used by the software architecture /model execution domain. The modeled domains have to store and retrieve Transfer Vectors as an opaque data structure, and in addition, provide the event and instance ids required to effect a send of the associated modeled event. These can be readily done.



Scope and Summary

Scope

The scope of this presentation was to assess the data handling required in the first step of a synchronous bridge scenario. That is, the invocation of the wormhole action in the HOME domain. In that effort it:

- Improved the B&W Information Model for clarity of presentation and use
- Walked through the 3 prescribed steps for constructing a bridge
 - Considered how each step could be performed
 - Introduced different types of data handling required based on the type of data item in the bridge interface
 - Considered how each type could be handled.

Summary

The data handling approach proposed by the B&W paper appears feasible for this first step of a synchronous bridge scenario with the exceptions of addressing 1) semantic conversion of Simple Value data items and 2) maintaining identifier mappings “half tables” when the instance population is dynamic.

Disclaimer



That said... the details of marshaling the identifier, “simple” and return mechanism data items can indeed be devilishly complex in the real world.

But it is feasible!

Questions? Comments?? Suggestions???

